

Human Ability

Lecture 5

Tim RMK GIGA

Departemen Teknik Informatika

Outline

[1] Visual Perception in Human

[2] Auditory Perception in Humans

[3] Tactile Perception in Humans

[4] Memory

Introduction

- Human computer interaction starts with human.
- Understanding the human is crucial.
- In this lesson, we are going to learn some **psychology** of what humans can do.

The Three Systems of Interaction in Humans

1. Input

- How stimuli are sent from the world and perceived inside the mind.

2. Processing

- Cognition - how the brain stores and reasons over the input it receives.

3. Output

- How the brain controls the individual's actions in the world.

Note: it is recommended to learn more about psychology especially the one focusing on sensation and perception.

Information Processing

- Humans consume input and produce output
- Humans act based on their perceptions.
- Here, we will look at how a person makes sense of input and then act in the world.
- We're not focusing on the source of the input and the impact of actions in the world



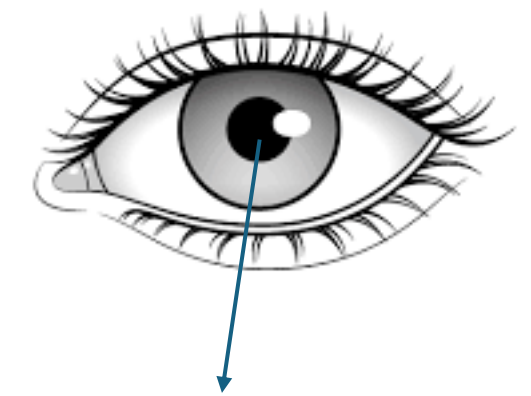
Visual Perception in Humans

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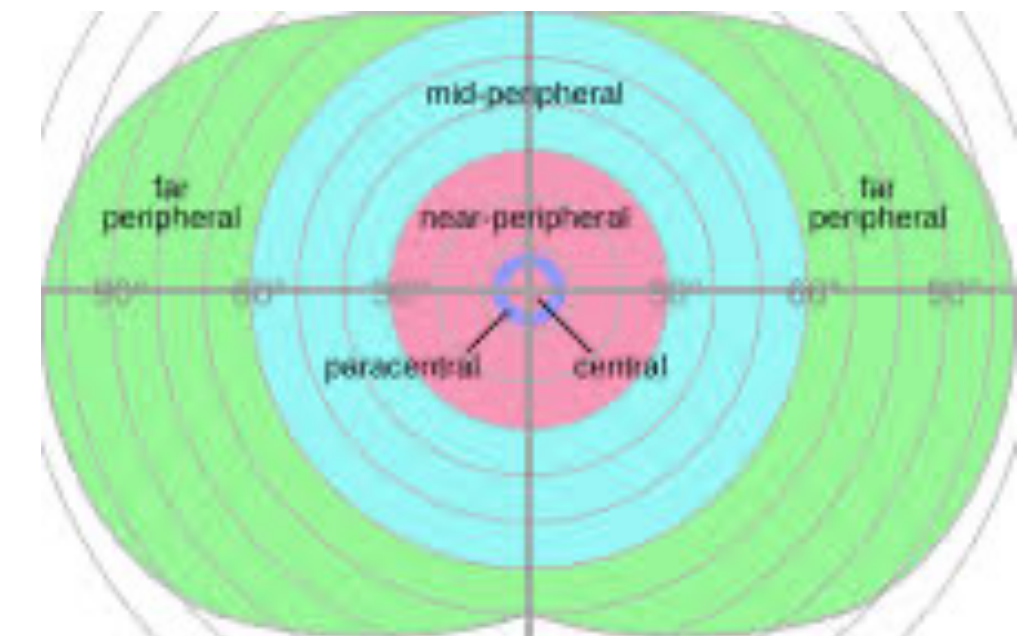
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Human's Visual System

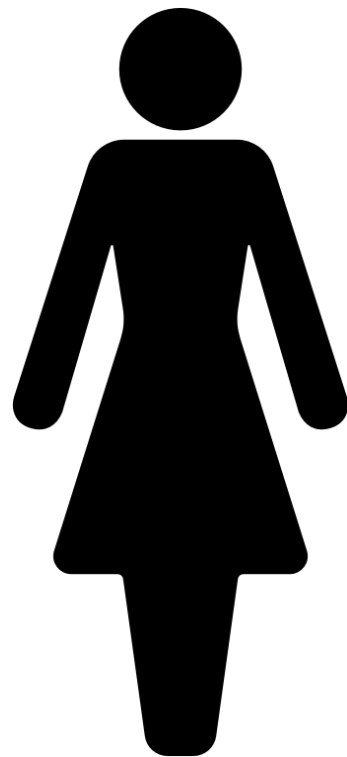
- Visual system is crucial for cognition. Majority of HCI concepts relate to visual perception.
- Humans have eyes. The eyes are useful for various tasks.
- The center of eye is most useful for focusing on color and tracking movement. Hence, important details should be placed in the center of human's view.
- Human's peripheral vision is good for detecting motion but not detail or color, so we use periphery for alerts.
- Sight is directional; missing visual feedback if looking the wrong way or eyes closed.



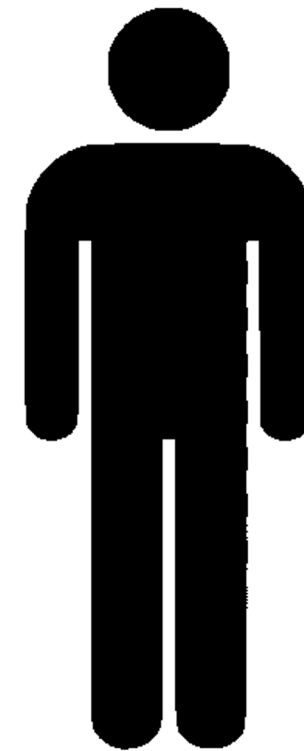
center of eye



Color Blindes Statistics



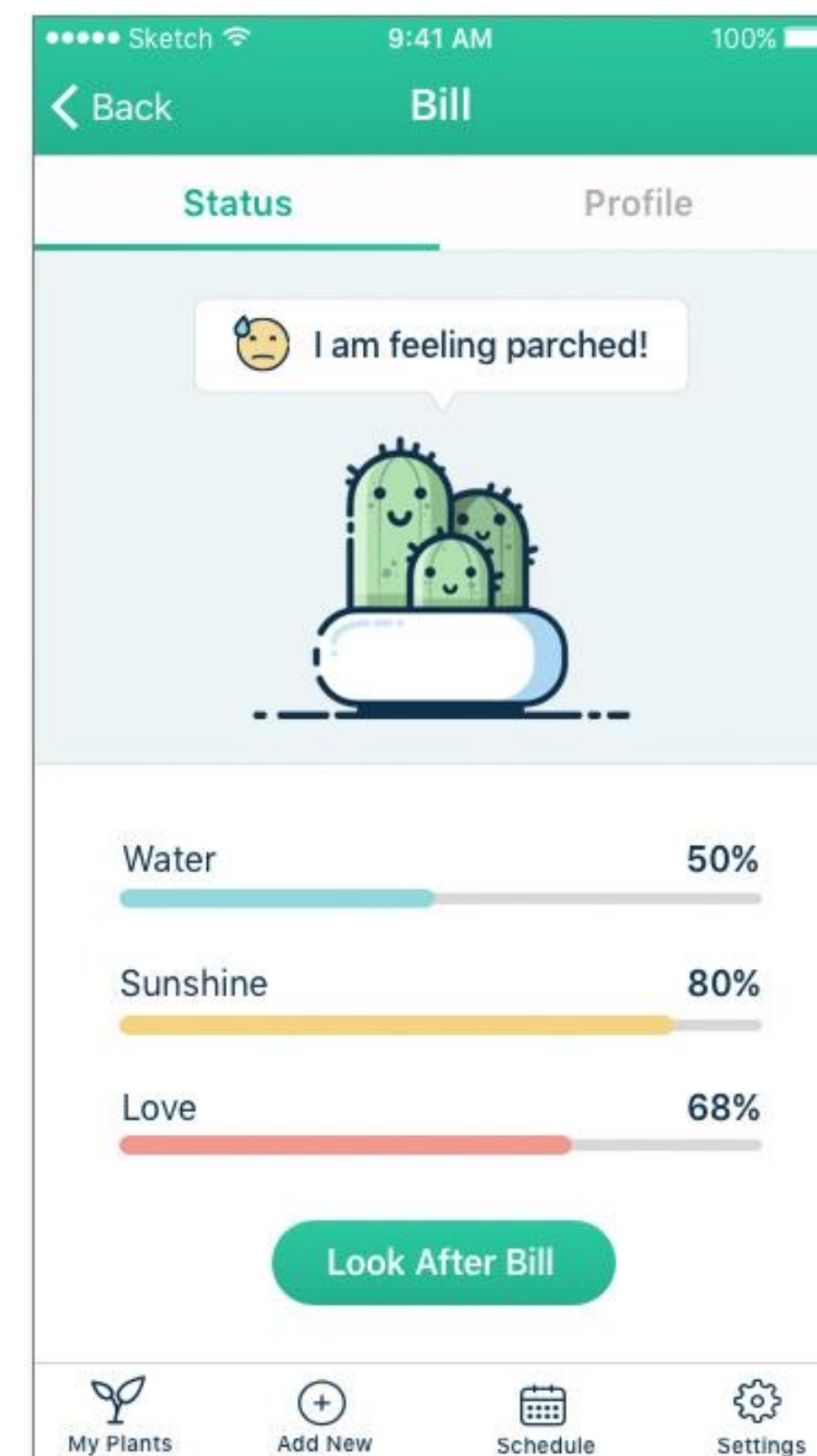
1 in 200 chance of
color blindness



1 in 12 chance of
color blindness

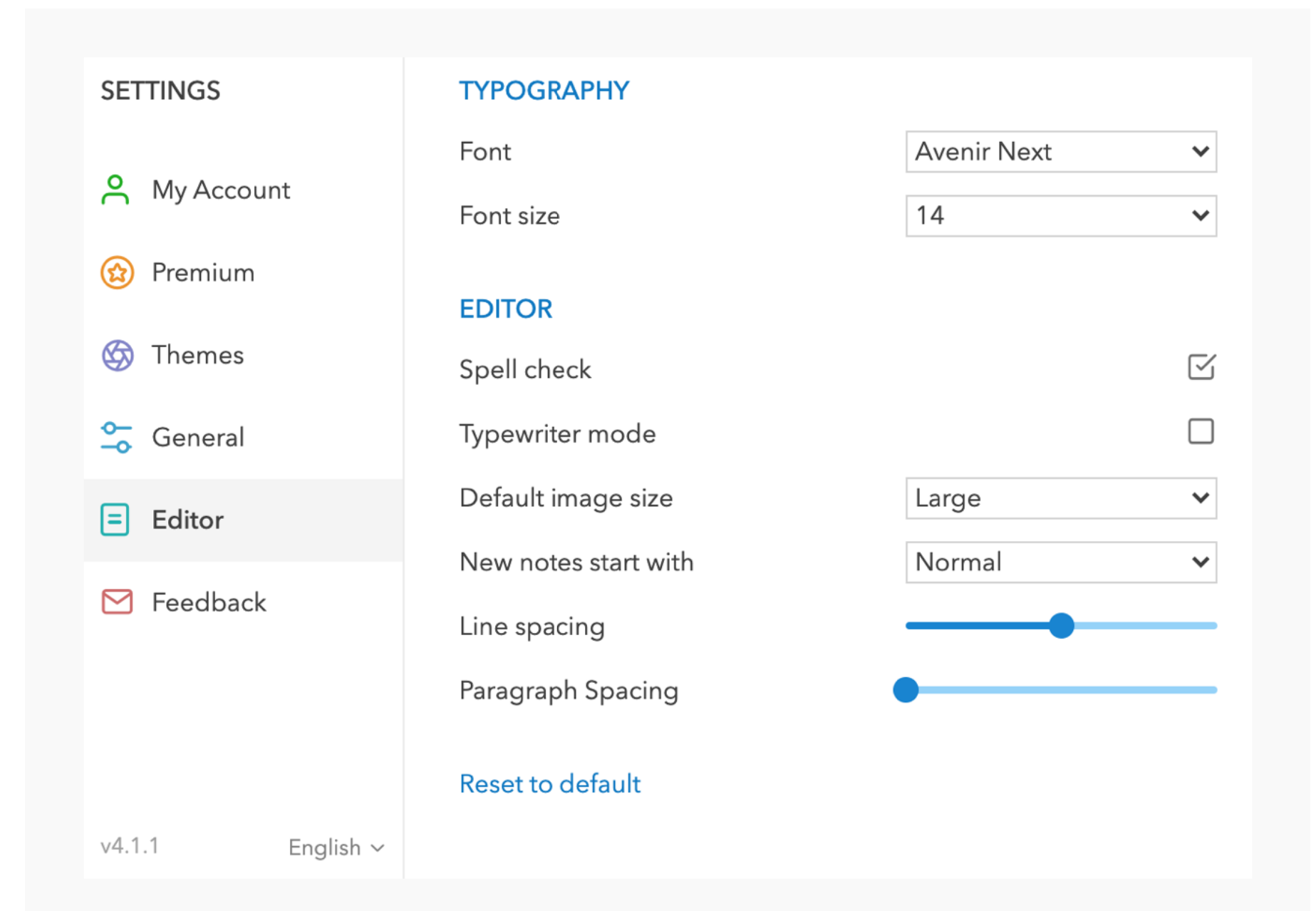
Design Considerations

- Avoid relying on color for understanding the interface.
- Use color to emphasize existing knowledge without depending on it.



Aging and Vision

- Aging Effects on Visual Acuity
- Visual acuity decreases with age.
- **Considerations for Design:**
 - Ensure font size is adjustable for older audiences.



provide setting to adjust font size



Auditory Perception in Humans

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Human's Auditory System

- Humans have remarkable hearing capabilities.
- Humans can discern noises based on pitch and loudness.
- Human ears also good at localizing sounds. They can distinguish between nearby quiet sounds and far away loud sounds even with similar pitches and loudness.
- Human hearing is not directional. They cannot close their ears.

Filtering Auditory Information

- As human hearing is not directional. Human face difficulty in filtering auditory information that useful for designing alerts.
- Problematic if overwhelming with too much information.
- Example:
 - When a person is typing it sounds click
 - When a person is typing it sound the letter he/she types
- **Note: Balancing alertness and information overload is crucial in design.**



Tactile Perception in Humans

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Introduction to Touch

- Human's skin can feel things upon contact.
- Characteristics of Touch
 - Can feel when things touch right against the skin.
 - Sensations include pressure, vibration, and temperature.
- Filtering Touch Feedback
 - Like listening, human can't easily filter out touch feedback.
 - Touch feedback is generally personal to the individual experiencing it.

Personal Feedback through Touch

- Touch can create more personal feedback compared to auditory feedback.
- Traditional touch feedback has been very natural.
 - **Example:** We feels the keys go down as she presses them on a keyboard.
- In modern touch technology like touchscreens, motion controls, and virtual reality, touch feedback needs to be designed explicitly into these systems.

Design Challenge: Message Alerts

- Problem: we want to be alerted when receiving a text message.
- The design constraints:
 - The phone must alert us whether the phone is in our pocket or on the table.
 - The alert must not disturb the people around us.
- Problem: vibrating loudly against the table counts as disturbing the people around us.
- Note: the alert not restricted to just one modality but are restricted to the sensors that the phone has available.



Message Alert Design - Solution

- Our smartphones have cameras and light sensors.
- We can use these to determine where the phone is and what kind of alert it should trigger.
- Example:
 - If the sensor detects light, that means the screen is visible, so it might alert us simply by flashing its flashlight or illuminating the screen.
 - If the sensor does not detect light, it would infer that the phone is in our pocket and thus, would vibrate instead.





Memory

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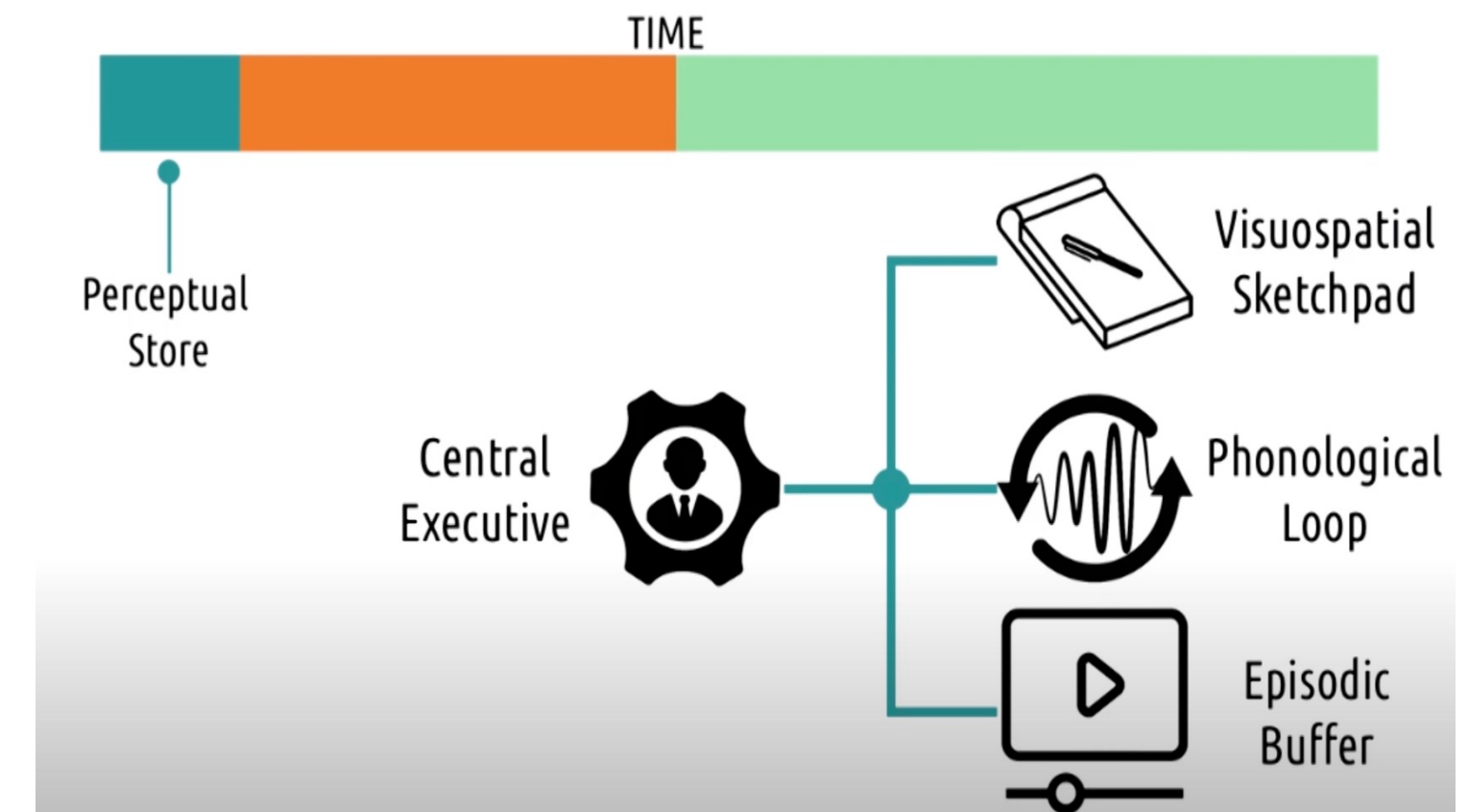
Memory: Perceptual Store



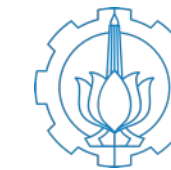
- There are various models of memory.
- Three primary types of memory:
 - Perceptual store (Working memory)
 - Short-term memory
 - Long-term memory

The Perceptual Store (Working Memory)

- Perceptual store is a very short-term memory lasting less than a second.
- One of the most common models of working memory came from Baddeley and Hitch in 1974.
- They described that working memory having three parts.
 1. **Visuospatial Sketchpad:** Holds visual information for active manipulation.
 - Example: picture a pencil. The visuospatial sketchpad is where you're currently seeing that pencil.
 2. **Phonological Loop:** Stores verbal/auditory information.
 - Example: Sounds or speech recently heard.
 3. **The Episodic Buffer:** Integrates information from other systems and maintains chronological ordering. Essential for forming coherent memories.
- All of these parts are coordinated by a central executive.



Activity: See this picture for 1 second



What was the score on the scoreboard?



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Reality

- Information in the perceptual buffer decayed very quickly. Thus, recalling images shortly after they are perceived is challenging.
- Fans of baseball may recall the score more accurately due to prior knowledge.
- Familiarity with the domain results in quicker and more accurate processing.
- Conclusion: Expertise allows for improved recall.

Memory: Short-Term and Chunking

- When designing interfaces, short-term memory is the biggest concern.
- We should not ask the user to remember too much at once.
- Current research shows that users can only store four to five chunks of information at a time.
- "Chunks" are units of information that our brain can process and store at one time.
- Imagine you're trying to remember a shopping list. If the list has only four or five items, like "milk, bread, eggs, cheese, and apples," you can probably remember all of them without writing them down. But if the list has ten items, like "milk, bread, eggs, cheese, apples, oranges, chicken, rice, pasta, and tomatoes," it becomes much harder to remember everything at once.

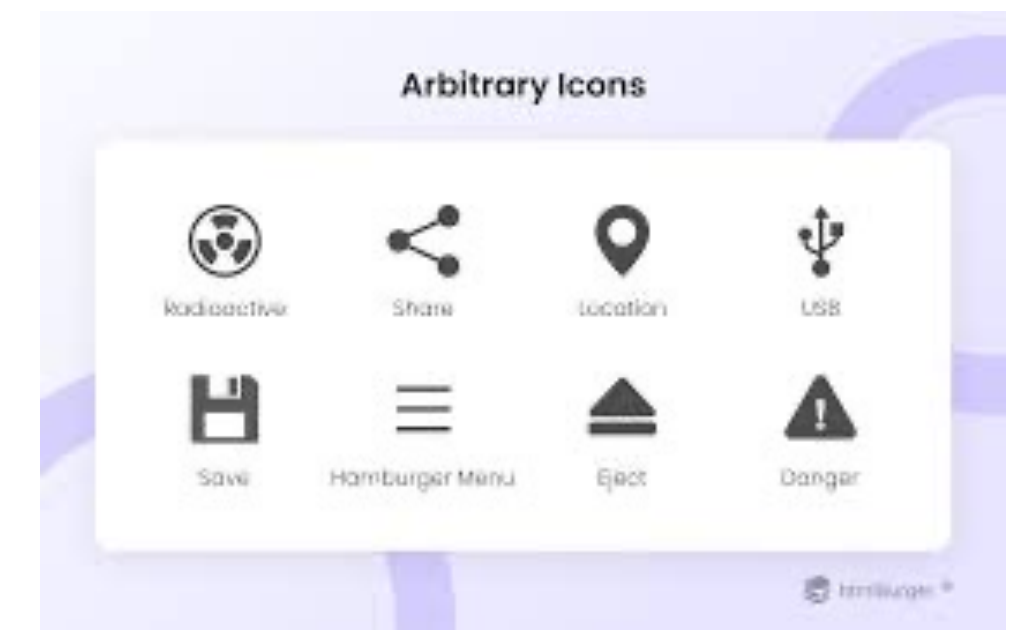
Memory: Short-term and Recognition

- Which of those words is easy to remember?
- Left-hand series: Involves memorizing five individual characters. Requires storing five chunks, making it harder to recall.
- Right-hand series can be grouped into word. Easier to recognize familiar items than to recall them from memory.

● FSDVW	● SOMIP	● PUPPY
● RTPVP	● REXIC	● TABLE

Implications for Interface Design

- Designing interfaces that leverage recognition can improve usability.
- Users can navigate more effectively when they recognize elements rather than having to recall them.
- Strategies to Minimize Memory Load:
 - Use familiar symbols and terminologies to facilitate recognition.
 - Create repetitive visuals and cues to enhance familiarity.



Enhancing Memory Capacity

- Chunking helps to shrink information into manageable units.
- Recommended memory load is four to five chunks.
- Example:

8 5 3 3 7 1 6 2 2 7 → 853 371 6227

Holding ten numbers may be too much.

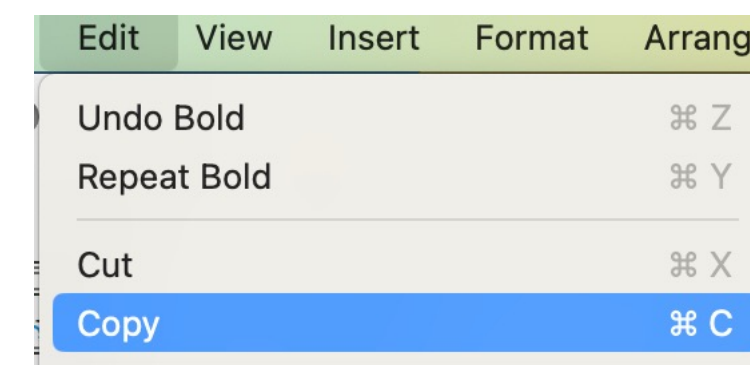
Familiar formatting enhances memory

Memory: Long-Term Memory

- Definition: Long-term memory is described as a seemingly unlimited store of memories, but it is more challenging to transfer information into long-term memory compared to short-term memory.
- Process of Storing Memories: To successfully load information into long-term memory, it typically needs to be rehearsed and stored in short-term memory multiple times.
- Example process of transferring information from short-term to long-term memory is a method of memorizing key-value pairs (e.g., Countries and capitals). The user focuses on the items they struggle to remember, repeatedly reviewing them to reinforce their memory over time.

Cognition: Learning

- Learning is a crucial cognitive process in interface design.
- "Cognitive" refers to the mental processes involved in gaining knowledge and understanding. These processes include thinking, knowing, remembering, judging, and problem-solving.
- When designing interfaces, the goal is to minimize the amount of learning required for users to find the interface useful while also teaching them to use it efficiently over time.
- Example of PowerPoint: this application helps new users learn by providing access to features (like the **Copy button**) while also **showing shortcuts** (hotkeys) to enhance efficiency.



Types of Learning

- Two primary types of learning:
 - Procedural Learning:** Learning how to do tasks. It is generally what we do in HCI. Example: using software, playing sports
 - Declarative Learning:** Knowledge about facts and information. Example: knowing the hotkey for paste

Procedural Learning



How to
do something

Declarative Learning



Knowledge
about something

Implicit vs. Explicit Knowledge

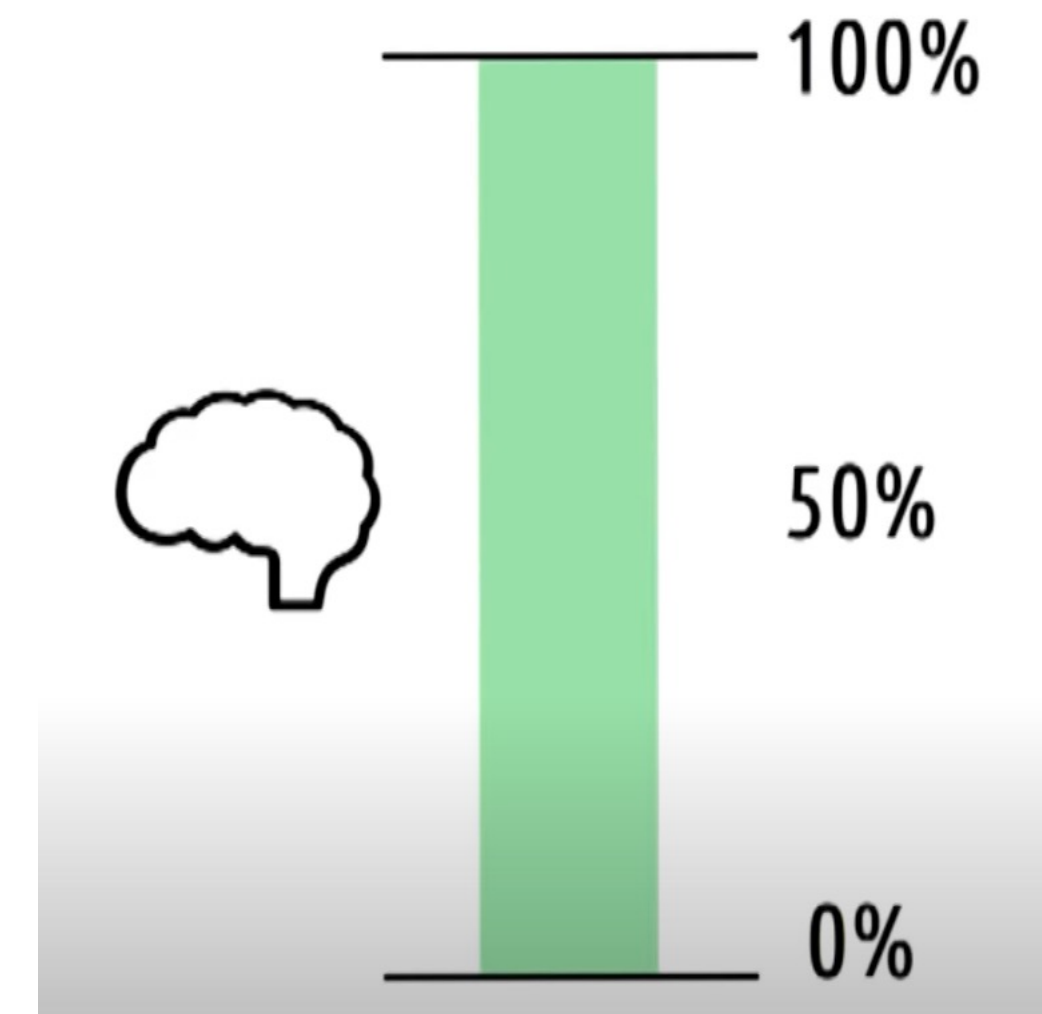
- Procedural knowledge often feels automatic (unconscious competence).
- Declarative knowledge is how we typically communicate with others.
- **The challenge of communication:** experts may find it challenging to explain their knowledge to novices because their procedural knowledge is often unconscious and second nature.

Importance in Design

- As designers, understanding both knowledge types is crucial to create user-friendly designs that cater to both novice and expert users.
- Designers must avoid creating interfaces suited only for their own expertise. They must balance the complexity and usability for diverse user levels.

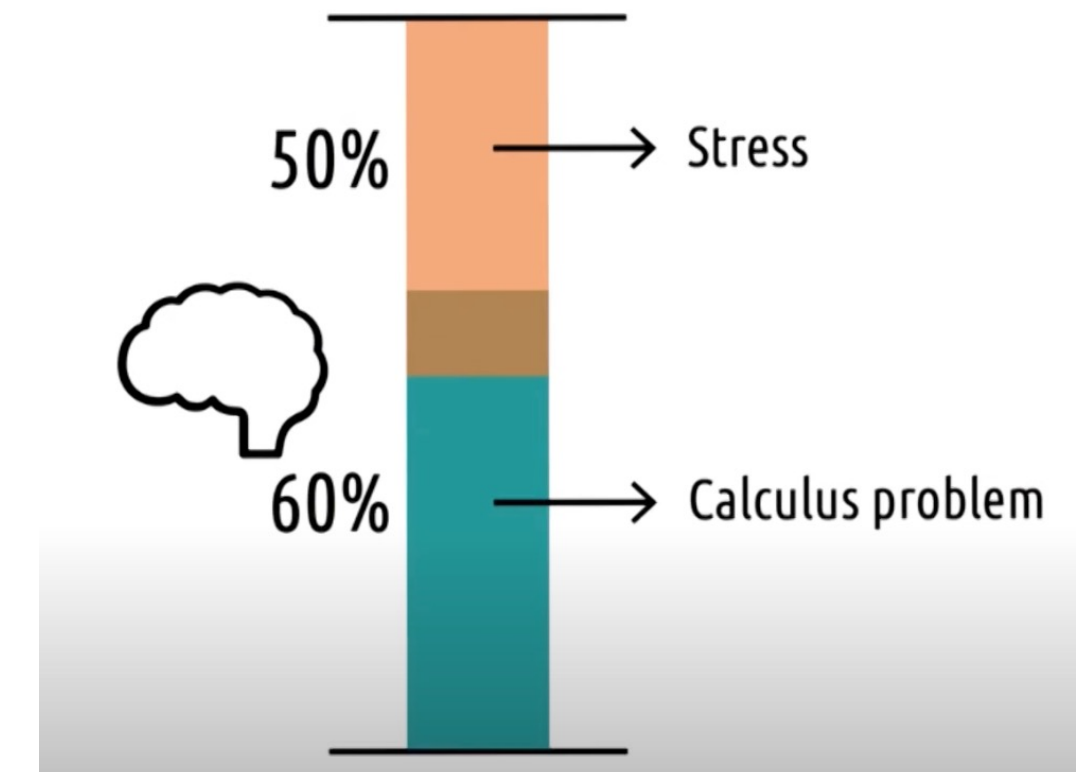
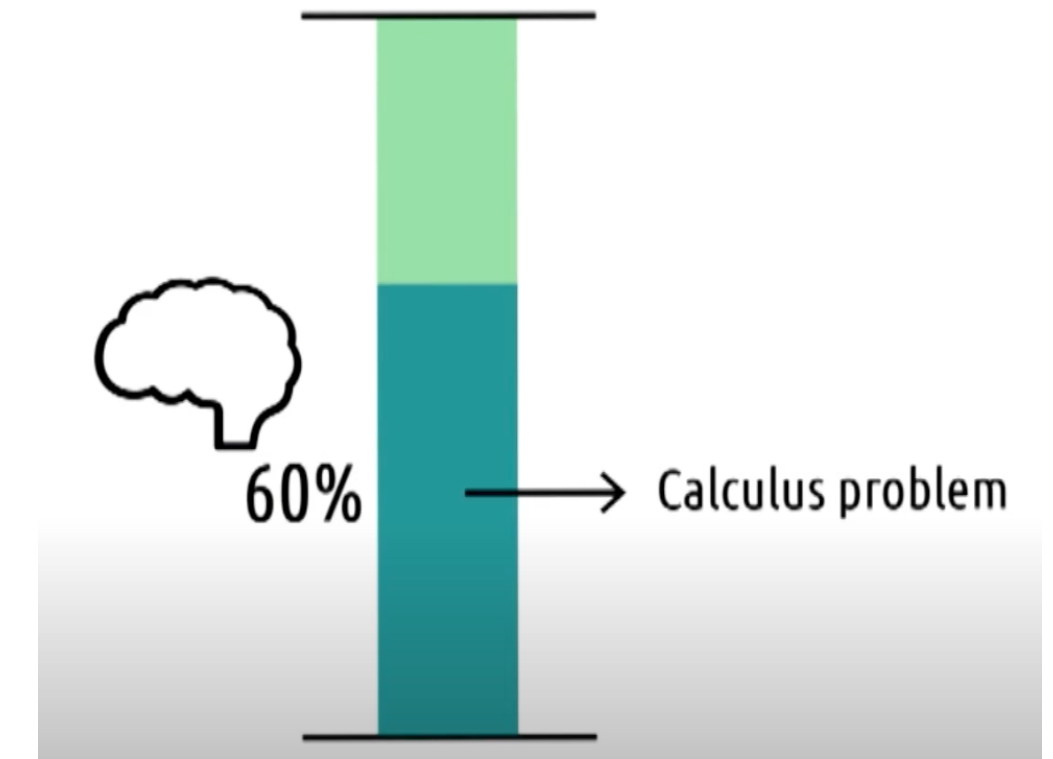
Understanding Cognitive Load

- Think of the brain like a computer.
- The brain has limited resources, similar to a computer's processor.
- Each task uses some of these resources.



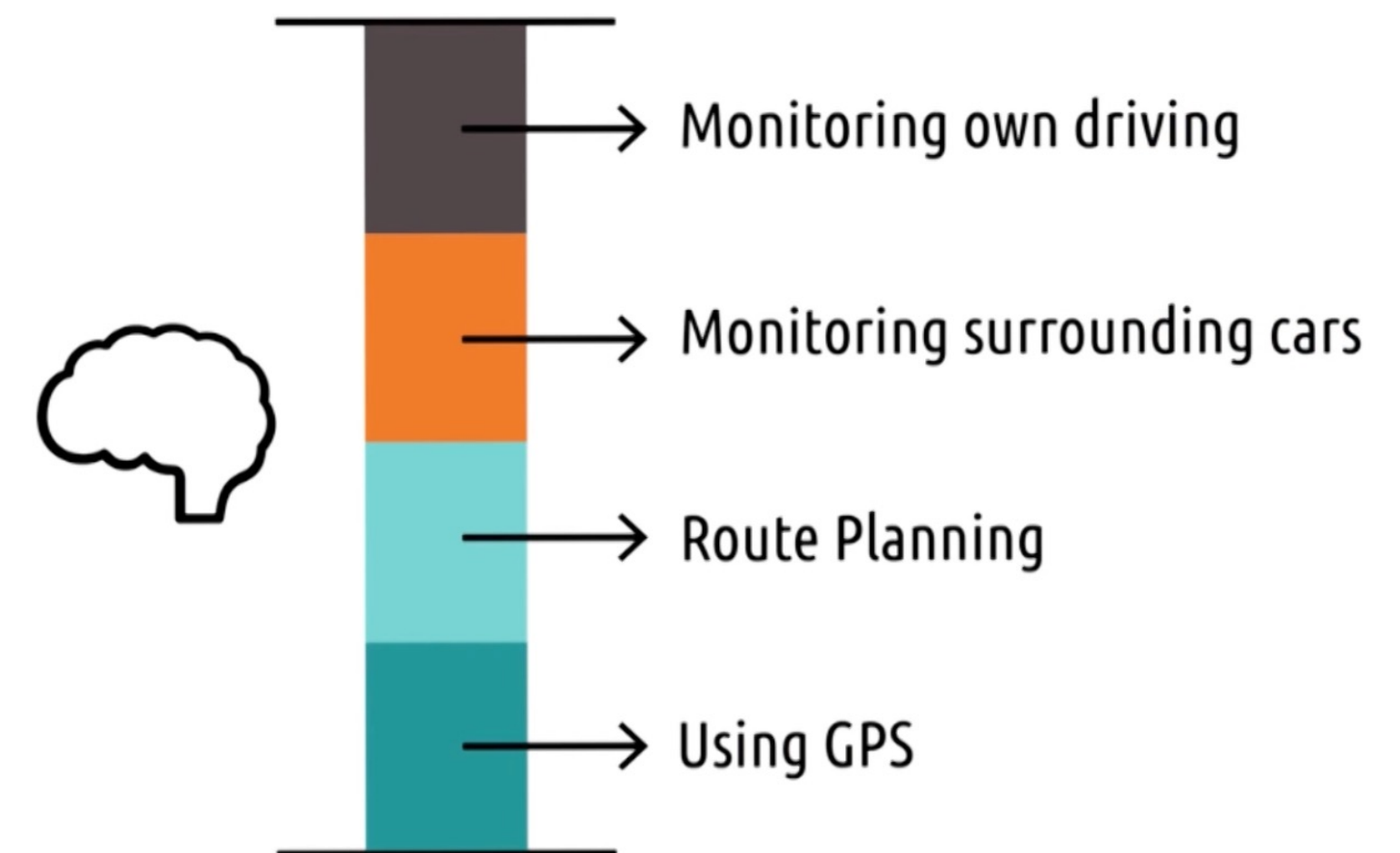
Example of Cognitive Load

- At home, solving a calculus problem uses 60% of your cognitive resources. In this setting, we have a plenty of resources to solve the problem
- If we go to a test and become stress. Stress during a test can take up 50% of cognitive resources.
- Less resources are available for solving problems. That's why test taking anxiety can have such a negative effect..



Applications of Cognitive Load in Design

- In HCI, the presence of distractions and anxiety disorders reducing user ability to do task.
- Important for designing user interfaces: Reduce cognitive load posed by the interface.
- Allow users to focus on the task.
- Example: GPS Systems
 - GPS systems need to account for users' limited cognitive resources because they're focusing on so many things at once



Five Tips for Reducing Cognitive Load in Interfaces

1

Tip #1 - Use Multiple Modalities

- Use both visual and verbal systems.
- Present information visually and verbally to prevent overload.

2

Tip #2 - Let the modalities complement each other

- Allow modalities to support each other.
- Avoid presenting conflicting information in different modalities.

3

Tip #3 - Give the user control of the pace

- Give users control over the pace of interactions.
- Avoid built-in timers that increase stress and cognitive load like disappearing menus.

Five Tips for Reducing Cognitive Load in Interfaces

4

Tip #4 - Emphasize essential content and minimize clutter

- Design interfaces that highlight important functions without overwhelming users.
- The principle of discoverability says we want the user to be able to find the functions available to them. But that could also raise cognitive load if we just give users a list of 500 different options.

5

Tip #5 - Offload Tasks

- Look closely at what the user has to do or remember at every stage of the interfaces operation and ask if you can offload part of that task onto the interface.

- **Examples:**

- Show previously entered information to aid recall.
- Automate tasks where possible to lessen user burden.



Thank You

Tim RMK GIGA

